Javad Komijani

Curriculum Vitae

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Professional Summary

Computational physicist and machine-learning researcher specializing in lattice gauge theory and scientific machine learning. Experienced in developing GPU-accelerated machine learning frameworks and differentiable computing tools using PyTorch. Strong background in mathematical physics, effective field theories, statistical modeling, and Monte Carlo methods.
Research focus: symmetry-aware (equivariant) machine learning for high-energy physics.

Education

2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*
- Developed theoretical models based on effective field theories.
- Built Bayesian inference pipelines for extracting physical parameters from large-scale lattice-QCD simulations using the developed theoretical models.
- Determined decay constants of D mesons, using lattice-QCD simulations and Bayesian analysis.
- Developed a novel definition for eigenvalues in the context of nonlinear problems.

2009 **MSc in Electrical Engineering**, *University of Tehran*
- Focus: electromagnetic waves and fields, signal processing, statistical analysis.

2006 **BSc in Electrical Engineering**, *University of Tehran*

Research Experience

2021–2026 **Postdoctoral Researcher**, *ETH Zurich*
- Developed a novel score-matching framework for diffusion models on non-Abelian Lie groups for lattice gauge theory, enabling generative modeling of SU(N) gauge configurations.
- Implemented distributed training infrastructure for large-scale lattice simulations.
- Built installable PyTorch extensions for differentiable linear algebra and adjoint-based ODE solvers supporting Lie-group-valued systems.
- Contributed machine-learning modules to the EU interTwin Digital Twin Engine, integrating normalizing flows for lattice simulations.
- Supervised PhD and Master's students in scientific machine learning and computational physics.
- Collaborated on international projects in scientific machine learning and high-energy physics.

2019–2020 **Postdoctoral Researcher**, *University of Tehran*
- Applied lattice simulations and Bayesian analysis to determine Standard Model parameters.

2017–2018 **Postdoctoral Researcher**, *University of Glasgow*
- Focused on determination of various form factors and quark condensates, using lattice simulations and Bayesian analysis.

2015–2017 **Postdoctoral Researcher**, *Technical University of Munich*
- Focused on determination of quark masses and various decay constants, using lattice simulations and Bayesian analysis.

Scientific Software and ML Infrastructure

lattice_ml **PyTorch scientific software package**
- Diffusion-based generative modeling on Lie groups and gauge theories.
- Adjoint-based differentiable ODE solvers for scalar and Lie-group-valued systems.
- Differentiable SVD and eigendecomposition of unitary matrices.

normflow PyTorch software package for normalizing flows on Lie groups and lattice gauge theory.

interTwin Contributed normalizing-flow-based ML modules to the EU interTwin Digital Twin Engine for lattice QCD simulations.

Selected Projects

SU(N) Diffusion Extended score-matching diffusion models to Lie groups for lattice gauge theory and applied generative modeling of SU(3) gauge configurations in two and four dimensions.

Quark Masses Proposed and implemented a method to calculate heavy quark masses; led to precise determination of the up-, down-, strange-, charm-, and bottom-quark masses using the MILC highly improved staggered-quark ensembles.

MRS Mass Defined a novel quark mass scheme, the minimal-renormalon-subtracted (MRS) mass, which is free from the shortcomings of other quark masses and provides a more robust foundation for precise quark mass determination.

Eigenvalues Generalized eigenvalue problems to non-linear cases using properties of solutions to the Schrödinger equation.

Technical Skills

Programming	Python, C++, Cython, Bash.
ML/AI	PyTorch, Diffusion Models, Normalizing Flows, Score Matching, Distributed Training.
Mathematics	Differential Equations, Linear Algebra, Optimization, Statistical Modeling.
Tools	Linux, Git, SQL, Pandas.
Scientific	NumPy, SciPy, Scikit-learn, Monte Carlo Methods.

Supervision and Mentoring

Postdoc.
- Gaurav Sinha Ray (2023–2025) — ML modules for EU interTwin lattice QCD project.

PhD students
- Octavio Vega (2024–present) — Group-equivariant diffusion models for lattice field theory.
- Ben Izett (2026) — Fourier acceleration for SU(N) lattice gauge theory.

MSc students
- Lara Turgut (2025–2026) — Gauge-equivariant diffusion models for lattice field theory.
- Ahmed Chahlaoui (2024–2025) — Autoregressive models for lattice configurations.
- Antoine Misery (2024) — Continuous-time generative models for lattice fields.
- Flavia Giehr (2024) — Continuous normalizing flows with adjoint methods.
- Klemen Kersic (2023) — Rational-quadratic splines for normalizing flows.
- Lea Segner (2023, co-supervised with J. Pinto) — Simulated tempering for topological freezing.

BSc students
- Nicolas Müller (2025) — Path integral formulation of quantum mechanics and Monte Carlo Simulations.

Interns
- Elias Nyholm (2023) — CNN architectures in arbitrary dimensions.

Teaching Experience

ETH Zurich
- **Head teaching assistant; substitute lecturer:** Quantum Mechanics II (Spring 2024), Quantum Mechanics I (Fall 2024).
- **Proseminar mentoring and supervision:** General Relativity (Spring 2025), Riemann Surfaces in Mathematical Physics (Spring 2023), Systems Out of Equilibrium (Fall 2022), Quantum Information (Spring 2022), Theoretical Physics (Fall 2021).

UT
- **Organizer and lecturer:** Workshop on lattice field theory for students from several universities in Tehran (Fall 2019).

UG
- **Teaching assistant:** C Programming under Linux (Spring 2018).

TUM
- **Teaching assistant:** Elementary Particle Physics (Spring 2016) and Quantum Field Theory (Fall 2015).

Washington University in St. Louis
- **Teaching assistant:** Statistical Mechanics (Spring 2015), Quantum Mechanics (Fall 2013), Physics I (Fall 2012), Physics II (Spring 2011), Special Relativity (Spring 2011).

Language Skills

English Fluent
German Basic (A2)

Publications

Full list of peer-reviewed publications: available here.